Extending vitrivr-engine with Emotion-Based Retrieval and a Modular User Interface

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Abstract. Interactive video retrieval has traditionally focused on visual, textual, and audio cues. Thereby, the emotions contained within this multimedia content were mostly overlooked for retrieval. In this work, we introduce a new version of the vitrivr-engine that incorporates emotion-based retrieval as a novel modality. It extends the established approaches such as visual concept detection, optical character recognition (OCR), and automatic speech recognition (ASR). To achieve this, we integrate deep learning models for facial expression analysis, text-based sentiment classification, and speech emotion recognition, each contributing to a unified representation of affective characteristics in video data. In addition to this new retrieval modality, we present vitrivr-web, a newly developed modular frontend built in React, designed to offer an adaptable and intuitive user experience through the modular structure of the vitrivr-engine. Furthermore, the backend features a new API, simplifying the use of the vitrivr engine and ensuring consistency across all functionalities. Together, these new features aim to expand the scope of interactive video retrieval, improving both usability and alignment with human memory processes.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Emotion-Based Retrieval

1 Introduction

Understanding and retrieving video content is not only about recognizing objects, scenes, or actions, but also about interpreting the emotions transmitted within the data. Emotions play a central role in how humans perceive and recall video content. Nevertheless, they remain a nearly underexplored dimension in interactive video retrieval systems. Incorporating affective signals into search can

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enable users to find content in ways that more closely align with human memory and perception. For instance, emotion-based search could help users locate emotionally charged moments, such as joyful reunions, tense confrontations, or scenes of sadness, within extensive video archives. Filtering results by affective content may improve relevance when users search for emotionally expressive material, such as uplifting clips or calm instructional content.

The Video Browser Showdown (VBS), an annual international competition, provides a challenging platform to evaluate such innovations in large-scale interactive video retrieval. VBS tasks require participants to locate known segments as quickly as possible, based on textual (KIS-T), visual (KIS-V) information, or by textual descriptions extended with questions asked by the participants (KIS-C). The ad-hoc video search (AVS), where an open description is provided, aims to find as many correct examples as possible, is the second query type. The last query type is the visual question answering (VQA), where manual inspection is needed to answer a specific question about a shown video. These tasks are performed on three different video datasets: the V3C dataset [17], which provides nearly 4,000 h of video content; the LHE75 dataset [12], containing 75 videos from laparoscopic gynecology; and the MVK dataset [19], offering more than 12 h of marine video content. This variety of content builds interesting conditions to test how and where emotion-based retrieval can complement traditional approaches, such as visual concept detection, speech analysis, and text-based search.

In this paper, we present the newest version of vitrivr [7], which introduces two key advancements. First, we integrate emotion detection as a novel retrieval modality, allowing users to issue queries or refine searches based on the affective characteristics of video segments. Second, we implement a modular user interface (UI) designed to support flexibility in retrieval workflows and tight coupling with the backend. The modular design enables individual components to be easily reconfigured or extended, providing an intuitive interaction experience for the modular backend. To fulfill this, we also reworked the API of vitrivr-engine.

2 Emotion-Based Retrieval

Emotions strongly influence how people interact with systems, shaping attention, decision-making, and memory. Incorporating emotion recognition into retrieval tasks enables more adaptive and context-aware responses, improving personalization and user satisfaction. Furthermore, emotion also plays a crucial role in video content. The emotional state of a video can be a key characteristic.

2.1 Emotion

Defining *emotion* and *sentiment* remains difficult across disciplines. Following Cabanac, we view emotion as *"any mental experience with high intensity and high hedonic content"* [4]. No universal taxonomy exists, though Ekman's six basic emotions: happiness, sadness, anger, fear, surprise, and disgust [5]. Other frameworks expand or modify these categories, resulting in variations in emotion recognition systems that depend on context and application.

2.2 Emotion Recognition

Emotion recognition seeks to identify human affective states from cues such as facial expressions, voice, or text [9]. In computing, it supports applications in healthcare, marketing, and security [8]. This study focuses on three modalities: (i) *Facial Emotion Recognition (FER)* Deep learning methods have greatly advanced FER, often achieving near-perfect accuracy on benchmarks like CK+ [3,10]. However, performance in real-world settings remains challenged by lighting, spontaneous expressions, and other uncontrolled factors [14]. (ii) *Text-based emotion classification* identifies emotional states from linguistic patterns in written communication. With the growth of social media and digital messaging, this modality has become increasingly important. Methods range from lexicon-based approaches to transformer architectures like BERT and RoBERTa, which model contextual language features and achieve strong performance on benchmarks such as ISEAR [1]. Key challenges remain, including ambiguity, informal language, and limited transferability across domains [2]. (iii) *Speech-based classification* uses acoustic and prosodic features to infer emotional states from voice signals. Conventional techniques relied on manually composed features, whereas newer classifiers integrate deep learning architectures (CNNs, RNNs, and attention mechanisms). Complex temporal patterns can therefore be modeled in a better way. Despite improvement, challenges remain, such as dealing with speaker variability, different languages, and background noise. These can significantly affect performance in real-world applications [6,11].

2.3 Emotion-Based Retrieval

We define emotion-based retrieval as the ability to search and filter content according to its affective characteristics, whether those of the media itself or of the user interacting with the system. The system can detect and index emotions expressed within videos, allowing users to query for emotionally charged scenes directly. Furthermore, it can also adapt retrieval behavior to the user's current emotional state, for instance, by recommending calming or uplifting content. This dual perspective distinguishes emotion-based retrieval from conventional approaches that treat emotion as irrelevant to query semantics [20].

3 Implementation

In this section, we firstly introduce the new retrieval by emotion functionality, before going into the architecture of vitivr-engine and its new backend features, followed by an overview of the new vitrivr-engine frontend.

3.1 Backend

The backend is provided by vitrivr-engine. Its functionality is mainly realized through two runtimes, the *indexing* and the *query runtime*, shown in Fig. 1.

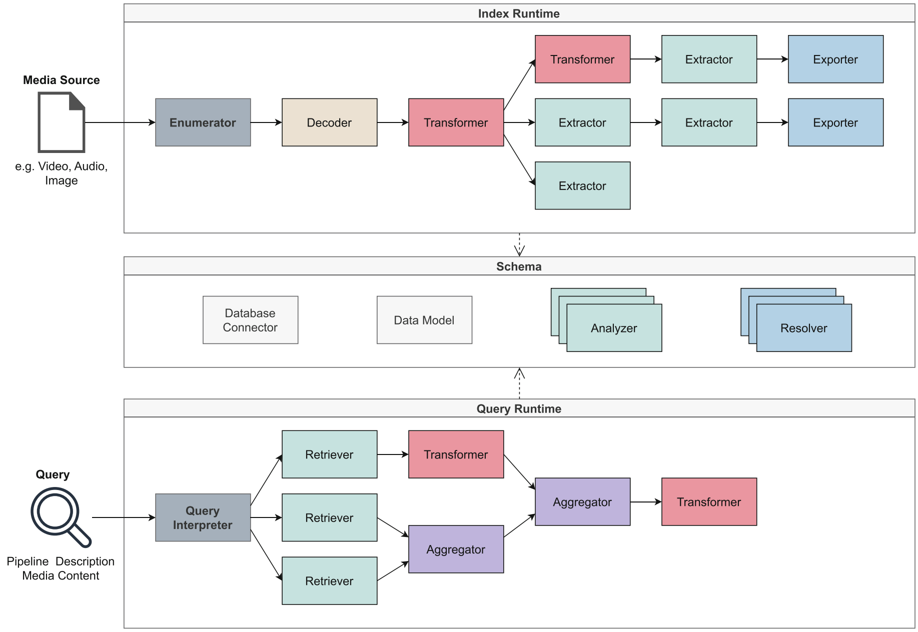

Fig. 1. The vitrivr-engine backend consisting of the *indexing runtime* and the *query runtime*. The vitrivr-engine allows formulating each indexing process and each query as a customized pipeline by concatenating operators that provide different functionalities.

In both runtimes, pipelines are composed from reusable operator types that implement common processing patterns. Decoders perform a $1:M$ expansion, for example by segmenting media items into smaller units suitable for further processing. Retrievers act as source operators that obtain stored features from the underlying database, while Extractors derive new features from media items and persist them back into the database. Transformers apply $1:1$ transformations, such as filtering or normalization, while Aggregators represent $M:N$ transformations, such as combining multiple elements. Exporters are responsible for persisting assets such as thumbnails to external storage systems, while access and data resolution for these storage operations is handled by a Resolver.

The flexible data model, in combination with a pluggable database abstraction layer, ensures consistent handling of media and features across both indexing and retrieval. The architecture is based on a flexible indexing model that enables the concatenation of different operator types, each providing distinct functionality. This makes it possible to apply various processing steps, such as temporal or spatial segmentation and the extraction of different feature sets, to multimodal *content* sources like videos, images, or audio. This flexibility allows vitrivr instances to be adapted easily to different application scenarios.

Queries in the system are expressed declaratively as sequences of composable operators, allowing for flexible and extensible retrieval workflows. This design separates the logic of query execution from the user interface. The query framework has been streamlined to emphasize clarity and consistency, reducing redundancy and making query pipelines easier to construct, understand, and maintain.

Finally, processing facilities are fully decoupled from public interfaces. vitrivr-engine can be embedded as a standalone library or exposed through a lightweight OpenAPI REST server. This separation ensures flexibility in integration with other projects and simplifies adaptation to diverse requirements. The current implementation is written in Kotlin and uses the PostgreSQL DBMS with the `pgvector` extension. Neural features such as OpenCLIP [15] and DINOv2 [13] are provided via external feature servers, while additional data sources such as transcripts and OCR can be seamlessly integrated.

3.2 Retrieval by Emotion

To incorporate emotion-based retrieval, we utilize three advanced models for emotion recognition across the various modalities. The VIT-Face-Expression model[1] analyzes facial expressions from the recognized faces using DeepFace [18] to discern emotional states. The emotion-text-classifier model[2] classifies textual input to find underlying emotions. Meanwhile, the wav2vec-english-speech-emotion-recognition model[3] captures emotional cues from audio signals.

Each of these models produces a vector representation of the detected emotions, which we store in our database. This method allows us to move beyond Boolean retrieval by emotion and enables more flexible retrieval through nearest neighbor search. Therefore, we avoid strict filtering by emotion. By querying the vector space, we can effectively retrieve content that matches specific emotional states or their combinations.

3.3 Frontend

The vitrivr frontend, vitrivr-web, was developed as a React application, as this framework enables straightforward modularization of components and facilitates deployment. The frontend architecture follows a modular, component-based approach, making it easily adaptable and extendible to accommodate potential changes in the backend. Queries are formulated through different components that act as building blocks. These components can either be used individually or combined to form temporal queries.

1 https://huggingface.co/trpakov/vit-face-expression.
2 https://huggingface.co/michellejieli/emotion_text_classifier.
3 https://huggingface.co/r-f/wav2vec-english-speech-emotion-recognition.

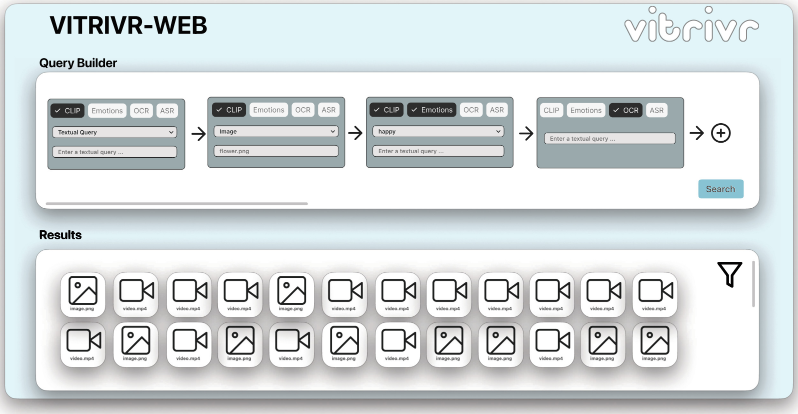

Fig. 2. Mockup of vitrivr-web.

Each block provides a selection for the modality type (CLIP, emotions, OCR, ASR). CLIP enables users to submit either textual queries or perform image-to-image searches to retrieve the k nearest neighbors. In the context of emotions, a textual query (CLIP) can be integrated with a filter to restrict results to a particular emotion. Emotion detection is a multimodal process that integrates facial expression analysis, automatic speech emotion recognition, and OCR-based text emotion analysis into a unified score. Furthermore, the system is equipped with the capability to process both OCR and ASR through textual queries.

The frontend further provides functionality to display and filter search results for faster access. The available filters encompass options such as media type, including videos, images, and 3D models. The ensuing results are presented in descending order of relevance. With regard to the video results, playback is initiated at the frame that most closely corresponds to the query. The filtering functionality is of particular importance in the context of video retrieval competitions, as it enables the refinement of the result set. Moreover, the frontend incorporates the DRES framework, which was developed to assess the performance of multimedia retrieval systems [16] Fig. 2.

4 Conclusion

In this paper, we present the latest iteration of the vitrivr-engine, which we plan to use in the VBS 2026 competition. This version builds upon the one that participated in the previous VBS competition, enhancing and refining its core functionalities. The three most significant advancements include the introduction of emotion-based retrieval functionality for the vitrivr-engine. The second significant change is the development of a dedicated, modular user interface designed to align with and leverage the system's modular architecture. Additionally, we have adapted the APIs to enhance interoperability and integration. With these developments, we aim to significantly improve the flexibility and retrieval capabilities of the vitrivr-engine in the upcoming evaluation campaign.

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